

# Learning the art

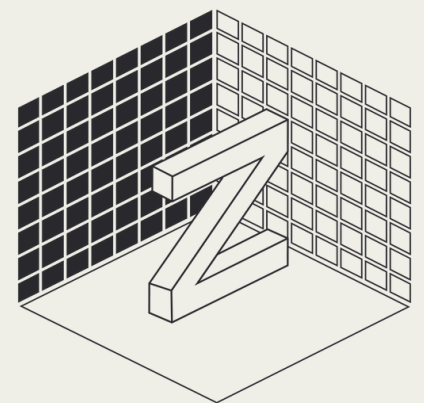
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OF FOSTERING OPEN-SOURCE COMMUNITIES

SciPy 2025

11th July, 2025

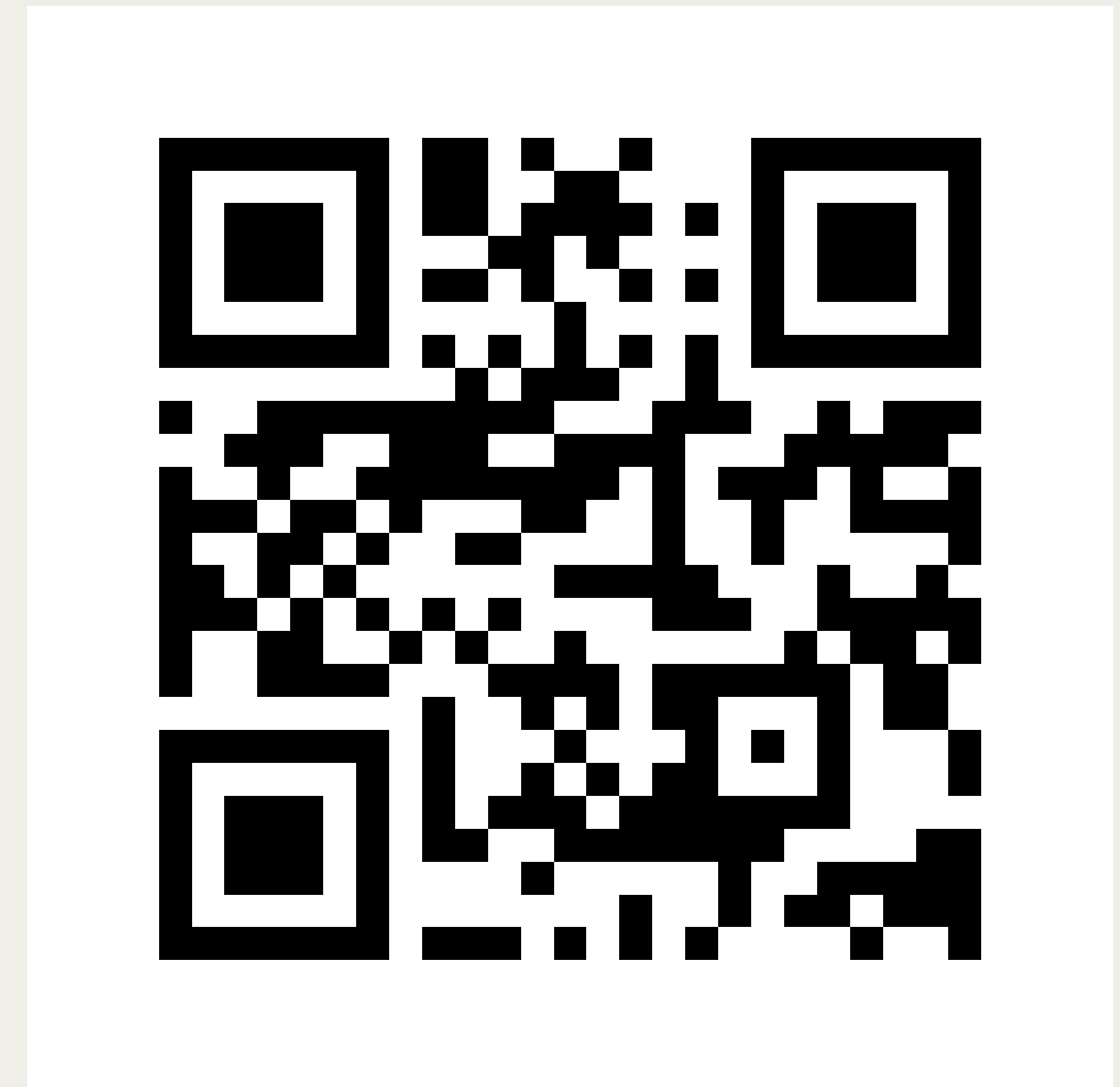
Tacoma, Washington



**Zarr**

# Sanket Verma

Software Engineer



<https://sanketverma.com/>

# Slides



[https://bit.ly/foster\\_community](https://bit.ly/foster_community)

# WHAT I'LL BE TALKING ABOUT?

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- But wait, \*what\* is the community?
- Best practices
- Pitfalls
- QnA

# Identifying the community for your open-source project



# WHAT ARE THE ELEMENTS OF THE COMMUNITY?

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Anyone who interacts with your project at any level; these could be —

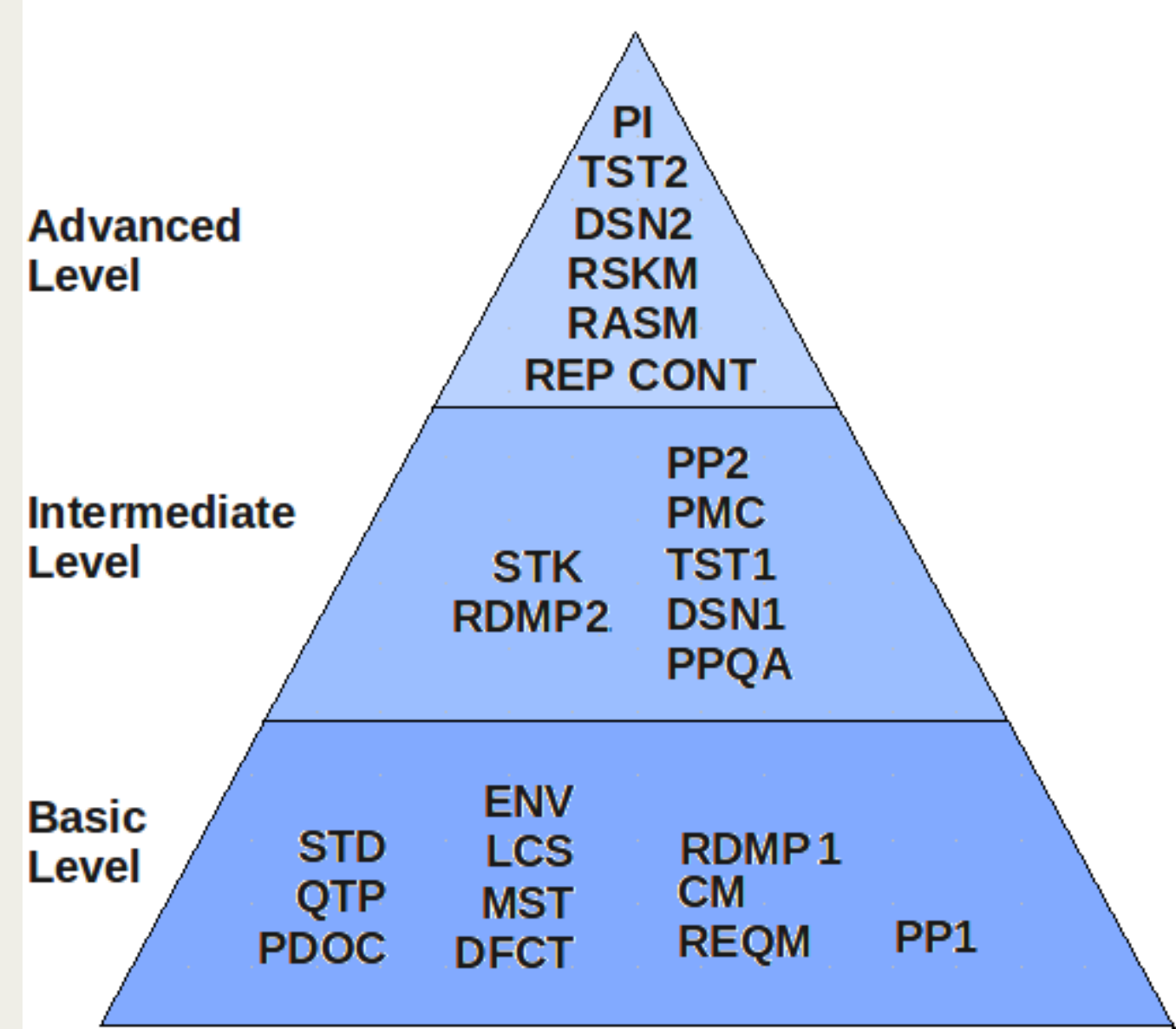
- Users
- Contributors
- Maintainers
- Governance team (mostly maintainers)
- Funders
- Organisations
- Research/academic institutions
- Advocates

How do I identify the  
maturity level of project?

# OPEN SOURCE MATURITY MODEL

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Provides a comprehensive methodology for the assessment —



Credits: [https://en.wikipedia.org/wiki/OpenSource\\_Maturity\\_Model](https://en.wikipedia.org/wiki/OpenSource_Maturity_Model)



# What do they actually mean?



# OPEN SOURCE MATURITY MODEL

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*Basic*

**Pre-Alpha, Alpha, Beta**

Basic documentation, Licenses and CoC, Technical Environment, Contributing guides, Tests, Commits and Bug reports

*Intermediate*

**Production/Stable**

Availability and use of the roadmap, design document, relationship with stakeholders, quality assurance, processes and workflows

*Advanced*

**Mature**

Reputation, contribution to OSS from organisations, product integration, funding

# Best practices to manage the community

# COMMUNITY MEETINGS

Ensure the meetings are public and open —

Today<>July 2025

Month

| MON<br>30 | TUE<br>1 Jul | WED<br>2                                                                   | THU<br>3                               | FRI<br>4                                                                     | SAT<br>5 | SUN<br>6 |
|-----------|--------------|----------------------------------------------------------------------------|----------------------------------------|------------------------------------------------------------------------------|----------|----------|
|           |              | <div><div></div>20:30 GeoZarr</div> <div><div></div>23:30 Zarr Offic</div> | <div><div></div>21:30 Zarr Steer</div> | <div><div></div>19:30 Zarr-Pyth</div>                                        |          |          |
| 7         | 8            | <div><div></div>23:30 Zarr Com</div>                                       | 10                                     | <div><div></div>19:30 Zarr-Pyth</div> <div><div></div>23:30 VirtualiZa</div> | 12       | 13       |
| 14        | 15           | <div><div></div>23:30 Zarr Offic</div>                                     | <div><div></div>21:30 Zarr Steer</div> | <div><div></div>19:30 Zarr-Pyth</div>                                        | 19       | 20       |
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Zarr Public Calendar

Events shown in time zone: (GMT+05:30) India Standard Time - Kolkata

[Add to Google Calendar](#)

<https://zarr.dev/community-calls/>

# How do I make them engaging?





# COMMUNITY MEETINGS

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## Crucial Updates and Pre-set Agenda

### Updates:

- Zarr participating in GSoC this year, ideas list: <https://github.com/numfocus/gsoc/blob/master/2025/ideas-list.md>
- ZEP9 Draft published — <https://zarr.dev/zeps/draft/ZEP0009.html>.
  - Follow up PRs in zarr-specs:
    - <https://github.com/zarr-developers/zarr-specs/pull/330>
    - <https://github.com/zarr-developers/zarr-specs/pull/331>
  - Setting up <https://zarr.dev/extensions>
- EP: Jackson lab will be converting the datasets to sharded Zarr V3 arrays
- SV: <https://2025.pycon.de/talks/ABWHSD/> — speaking at PyCon DE 2025!

### Open agenda (add here 📌):

- MS using Pizzarr for their work
  - GDAL and Pizzarr work for virtual references as well
  - Has datasets in HDF5 and NetCDF - both has their peculiarities
- JM: *gives an overview of ZEP9*
- EP: Jackson lab data conversion

<https://zarr.dev/community-calls/>

# COMMUNITY MEETINGS

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## Introductions w/ Ice-breakers

- Introductions
  - Josh: beverage of choice: whisky → gin.
  - Liam: finance → energy. forecasting things like Weather. slicing regularly grided. NZ so kiwi fruit juice
  - Eric: neuroscience/freelance. dirty chai latte.
  - Davis: big image datasets/freelance.
  - Altay: TGS lead data scientist. energy companies. wind, solar, gas. core is seismic. manage petabytes of data in zarr. Google cloud. (last 3 years). big fan. things slowing down. looking to help. coffee and old fashions.

# OFFICE HOURS

Hidden venue to understand and help your project community —

 **Zarr**  
chunked, compressed, N-dimensional arrays

Documentation

Contribute

Python Quickstart



Content

ABOUT

Description

Applications

Features

Sponsorship

Videos

SUBPAGES

Adopters

Blog

Community

Conventions

Datasets

Implementations

Office Hours

Slides

Specification

ZEPs

Zarr Office Hours

Join here

We're hosting office hours on Wednesdays every two weeks. Please join us if you have questions about Zarr and want to learn more about the storage format. We'll discuss the specification, the implementations, what's new in the Zarr ecosystem, how you can get involved in the community and much more.

Office hours are a great place to go if you want to start using Zarr and have questions about whether it suits your data storage needs. We'll have a few items on the agenda to kickstart the meeting, but the overall agenda and structure of the office hours will be shaped according to the attendees' and community's needs.

Please see the Zarr community calendar for exact timings:

Today

<

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July 2025

Month

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[Add to Google Calendar](#)



https://zarr.dev/office-hours/



# MANAGING ONLINE SPACES

Chat rooms, forums, mailing lists, etc.

Views

Recent conversations

Combined feed

Channels

[IG] Economics

Help

VirtualiZarr

Zarr

Zarrchive (Resolved Topics)

Log in to browse more

Recent conversations

Overview of ongoing conversations.

Search

Log in

Standard view

Include DMs

Unread

Participated

Filter topics (t)

| Channel | Topic                                           | Participants | Time         |
|---------|-------------------------------------------------|--------------|--------------|
| Zarr    | <a href="#">extensible dtypes</a>               | +2           | 13 hours ago |
| Zarr    | release 3.0.10                                  |              | 13 hours ago |
| Zarr    | release 3.1.0                                   |              | 15 hours ago |
| Zarr    | release 3.0.9                                   |              | 20 hours ago |
| Zarr    | local integration testing with pep 723 metadata |              | 21 hours ago |
| Zarr    | zipstorers                                      | +2           | 2 days ago   |

# Contributions vs. Conflicts



# CONTRIBUTIONS VS. CONFLICTS

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Ensure proper conflict resolution mechanisms are in place —

- Transparent decision-making process
- Code of Conduct and reporting mechanism
- Governance
- Documentation and Precedents

Ensure essential guides are updated and followed —

- Maintainers' guide
- Contributors' guide

# CONTRIBUTIONS VS. CONFLICTS

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Creating processes and workflows to handle substantial contributions —

## Zarr Enhancement Proposals (ZEPs)

Community Feedback Process for Zarr Specifications.

[Propose a new ZEP](#)

[View it on GitHub](#)

ZEP stands for Zarr Enhancement Proposal. A ZEP is a design document providing information to the Zarr community, describing a modification or enhancement of the [Zarr specifications](#), a new feature for its processes or environment. The ZEP should provide specific proposed changes to the Zarr specification and a narrative rationale for the specification changes.

We intend ZEPs to be the primary mechanism for evolving the spec, collecting community input on major issues and documenting the design decision that has gone into Zarr.

[active ZEPs](#) / ZEP0000

## ZEP 0 — Purpose and process

Author: Sanket Verma ([@MSanKeys963](#)), Zarr

Email address: [svsanketverma5@gmail.com](mailto:svsanketverma5@gmail.com)

Status: Active

Type: Process

Created: 2022-14-03

Discussion: <https://github.com/zarr-developers/governance/pull/16>

### What is ZEP?

ZEP stands for Zarr Enhancement Proposal. A ZEP is a design document providing information to the Zarr community, describing a modification or enhancement of the Zarr specification, a new feature for its processes or environment. The ZEP should provide specific proposed changes to the Zarr specification and a narrative rationale for the specification changes.

# CONTRIBUTIONS VS. CONFLICTS

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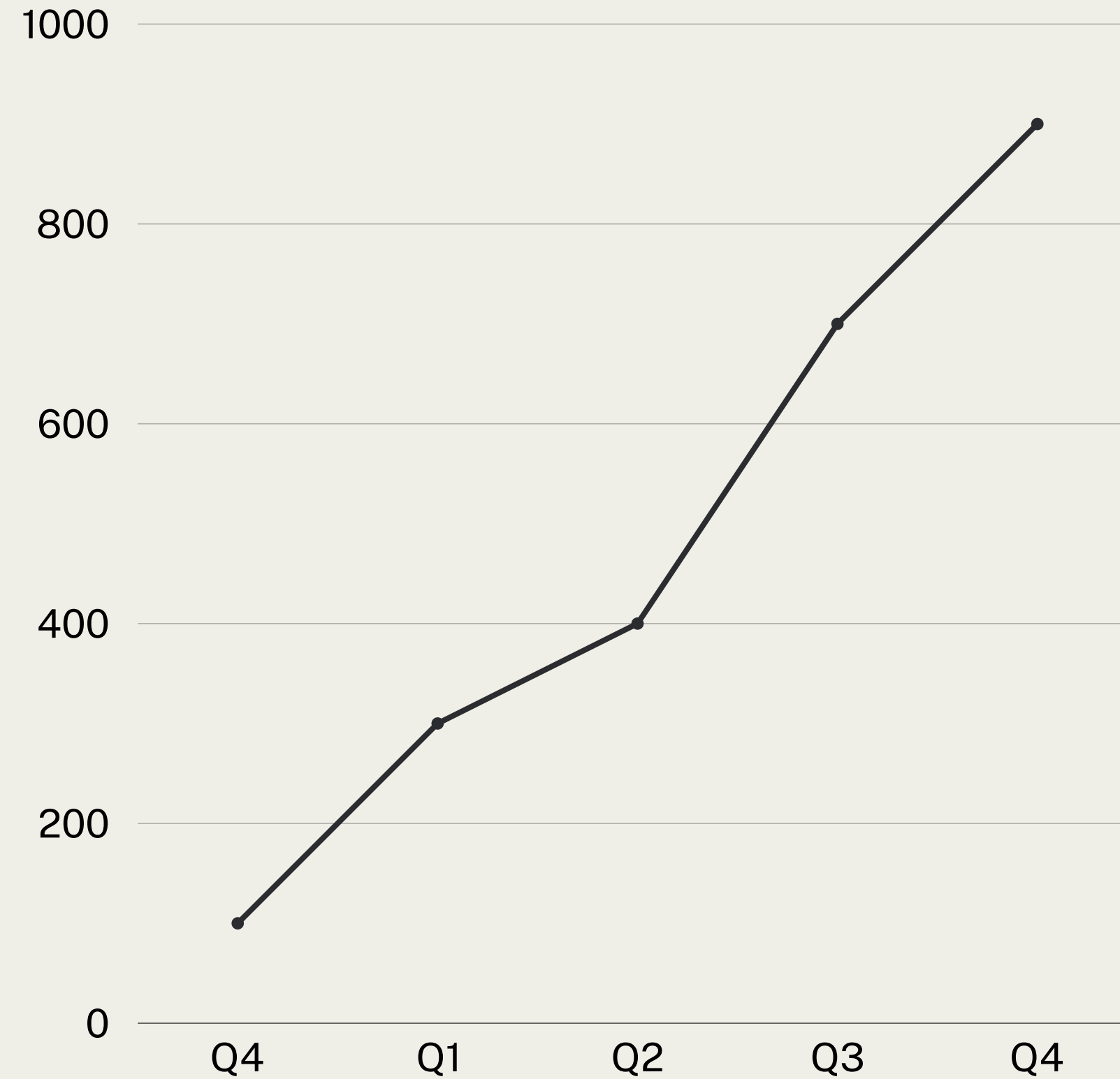
Braindump 🧠



<https://zarr.dev/blog/zep-inception/>



# How do I grow my community?



# HOW DO I GROW PROJECT'S COMMUNITY?

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Build scaffolding —

- Lowering the barrier of participation
- Be inclusive
- Streamline, attract and encourage contributions
- Participating in conference, sprints, hackathons

Identify important stakeholders —

- Potential donors
- Potential industrial adopters

Leverage educational programs —

- Google Summer of Code
- Outreachy

# Avoiding common pitfalls



# AVOIDING PITFALLS

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- Always iterate and improve
- Define clear pathways for leadership
- Clear offboarding process for inactive members
- CoC and a reporting mechanism
- Timely responses on issues and PRs help (hint: use a bot!)

# Thank you!

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**ANY QUESTIONS?**

Sanket Verma  
11th July, 2025

# Go back! ⬅

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NOTHING TO SEE HERE